9th Class • Punjab Board • PCTB New Syllabus

Chapter 3: The Cell

Solved notes in book order — MCQs, short answers, detailed answers, inquisitive questions and key points.

Chapter 3 • Units 3.1 – 3.5 • Exercise solved in book order

What this chapter is about

• The cell is the basic unit of life and the building block of every living organism. Robert Hooke discovered it in 1665 and Robert Brown observed the nucleus in 1831.

• The chapter describes every part of the cell: cell wall, cell membrane (fluid mosaic model), cytoplasm, nucleus, cytoskeleton, ribosomes, endoplasmic reticulum, Golgi apparatus, lysosomes, vacuoles, mitochondria, plastids, centrioles, cilia and flagella.

• After that it explains the structural advantages of plant and animal cells, cell specialization (mesophyll, epidermal, muscle, nerve, red blood, liver cells), division of labour and stem cells.

• The textbook exercise (14 MCQs, 14 short answers, 10 detailed answers and 2 inquisitive questions) is solved here in book order.

A. Multiple Choice Questions

Tick the correct option. The reason after each answer also tells you why the other options are wrong, which is useful for the paper.

1. The process of cellular respiration occurs in:

a) Nucleus
b) Mitochondria
c) Ribosomes
d) Golgi apparatus

Answer: b) Mitochondria — Mitochondria are the powerhouses of the cell; they break down glucose with oxygen (cellular respiration) and release energy as ATP.

2. The smooth endoplasmic reticulum (SER) is primarily involved in the synthesis of:

a) Proteins
b) Lipids
c) Carbohydrates
d) Nucleic acids

Answer: b) Lipids — The rough ER, which carries ribosomes, makes proteins; the smooth ER has no ribosomes and makes lipids, and also detoxifies harmful substances.

3. Ribosomes are composed of:

a) RNA and protein
b) DNA and protein
c) Carbohydrates and lipids
d) RNA and carbohydrates

Answer: a) RNA and protein — Ribosomes are made of ribosomal RNA (rRNA) and proteins; they have no DNA and are not covered by a membrane.

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4. What is the primary function of ribosomes?

a) Energy production
b) Protein synthesis
c) Lipid synthesis
d) DNA synthesis

Answer: b) Protein synthesis — Ribosomes are the sites of protein synthesis: they read the message of mRNA and join amino acids into a protein chain.

5. Which cell organelle is involved in packaging and modifying proteins?

a) Nucleus
b) Mitochondria
c) Golgi apparatus
d) Endoplasmic reticulum

Answer: c) Golgi apparatus — Proteins made on the rough ER are sent to the Golgi apparatus, which modifies, sorts, packages them into vesicles and secretes them.

6. Which cell organelle is responsible for breaking down waste materials?

a) Golgi apparatus
b) Nucleus
c) Mitochondria
d) Lysosome

Answer: d) Lysosome — Lysosomes contain strong digestive enzymes that break down waste materials, worn-out organelles and engulfed bacteria.

7. Which of the following cell structures is involved in maintaining cell shape?

a) Cytoskeleton
b) Centrioles
c) Nucleus
d) Lysosome

Answer: a) Cytoskeleton — The cytoskeleton is a network of microtubules, microfilaments and intermediate filaments; it holds organelles in place and maintains the shape of the cell.

8. Which specialized region of the nucleus is responsible for ribosome assembly?

a) Nucleoplasm
b) Nucleolus
c) Chromatin
d) Chromatin

Answer: b) Nucleolus — The nucleolus is the dense body inside the nucleus where ribosomal RNA is made and ribosomal subunits are assembled.

9. What is the main function of the nuclear pores?

a) Regulation of cell division
b) Control of pH of the cell
c) Protein synthesis
d) Control of transport of molecules

Answer: d) Control of transport of molecules — The nuclear envelope has many small pores through which proteins and RNA move between the nucleus and the cytoplasm.

10. Which of the following cellular structures is found in animal cells and helps in cell division?

a) Cell membrane
b) Centriole
c) Plasmodesma
d) Vacuole

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Answer: b) Centriole — At the start of cell division the pair of centrioles duplicates and forms spindle fibres that separate the chromosomes.

11. Which sub-cellular organelle plays a crucial role in energy production within the cell?

a) Endoplasmic reticulum
b) Golgi apparatus
c) Mitochondria
d) Lysosomes

Answer: c) Mitochondria — Mitochondria produce energy (ATP) by cellular respiration, so they are called the powerhouses of the cell.

12. In a multicellular plant, which cell type is responsible for the production of glucose?

a) Xylem
b) Phloem
c) Epidermal
d) Mesophyll

Answer: d) Mesophyll — Mesophyll cells are the green cells of the leaf; they contain many chloroplasts and carry out photosynthesis, which produces glucose.

13. Which organelle can double its number by itself?

a) Ribosomes
b) Lysosomes
c) Mitochondria
d) Golgi apparatus

Answer: c) Mitochondria — Mitochondria have their own DNA, so they can divide and increase their number by themselves.

14. Which of these are present on the surface of rough endoplasmic reticulum?

a) Ribosomes
b) Lysosomes
c) Mitochondria
d) Vacuoles

Answer: a) Ribosomes — The rough ER has ribosomes attached to its surface, which is why it looks rough and why it makes proteins.

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B. Write Short Answers

1. What are the main functions of cell membrane?

The cell membrane is a thin, living, partially permeable layer of lipids and proteins that covers the cell. Its functions are:

1. It keeps the cell contents in place and gives the cell its boundary.

2. It is selectively permeable, so it controls the movement of substances into and out of the cell.

3. It protects the cell from the outside environment.

4. It helps in cell recognition and communication, because of the proteins and carbohydrates present on its surface.

2. What key role does the Golgi apparatus play in eukaryotic cells?

The Golgi apparatus is a set of many flattened sacs (cisternae) stacked over each other. It receives proteins and lipids from the endoplasmic reticulum, modifies them, sorts them and packs them into vesicles. These vesicles are then secreted out of the cell or sent to other organelles. In this way the Golgi apparatus works as the "post office" of the cell, and it is also the organelle that forms lysosomes and secretory vesicles.

3. How do lysosomes contribute to the cell's functioning?

Lysosomes are single membrane-bound sacs filled with strong digestive enzymes. They:

1. break down waste materials, worn-out organelles and food particles into simple substances that the cell can use;

2. digest the bacteria and other harmful particles that the cell engulfs, so they act as the cell's defence force;

3. help in the digestion of food inside food vacuoles and remove the cell's waste. Because their enzymes can even digest the cell's own parts, lysosomes are called the "suicidal bags" of the cell.

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4. Which organelle detoxifies harmful substances and breaks down lipids?

Answer: Smooth Endoplasmic Reticulum (SER)

The SER has no ribosomes on its surface. It synthesizes lipids, stores calcium ions and neutralizes harmful (toxic) substances such as drugs and poisons, specially in the liver cells.

5. What is the smooth endoplasmic reticulum responsible for?

The smooth ER is responsible for: (i) synthesis of lipids and steroid hormones; (ii) detoxification of harmful substances; (iii) storage and release of calcium ions; and (iv) transport of these materials to other parts of the cell. It is called smooth because it has no ribosomes on its surface.

6. How do the vacuoles in plant cells differ from vacuoles in animal cells?

Vacuoles in plant cells	Vacuoles in animal cells
Most mature plant cells have a single, large, central vacuole.	Animal cells usually have many small, temporary vacuoles.
The membrane of the plant vacuole is called tonoplast and the fluid inside is called cell sap — a watery solution of salts and sugars.	They contain water and food substances; contractile vacuoles of amoeba pump out extra water and wastes.
The large vacuole pushes the nucleus to a side and creates turgor pressure that keeps the cell firm and maintains its shape.	They do not produce such turgor pressure, because they are small in size.

7. What could happen if lysosomal enzymes stop working properly?

If lysosomal enzymes stop working, the waste materials, worn-out organelles and engulfed bacteria will not be digested. These substances will collect inside the cell and make it diseased. Lipids and other materials will also store up in the tissues. Such problems are called lysosomal storage diseases. If the lysosomes leak their enzymes into the cytoplasm, they may digest the cell's own parts and the cell will die, which is why they are called suicidal bags.

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8. Why are the cristae important for cellular respiration?

Cristae are the inward folds of the inner membrane of a mitochondrion. They increase the surface area of the inner membrane, and the enzymes needed for cellular respiration are present on this membrane. Therefore more cristae mean more space for the respiratory enzymes and more energy (ATP) production. They also hold the respiratory chain in a small space with the matrix, so that the reactions of cellular respiration occur efficiently.

9. How are chromatin and chromosomes related?

Chromatin is the network of long thread-like fibres present in the nucleoplasm of the nucleus; it is made of DNA and proteins. When the cell divides, this chromatin coils and condenses (becomes shorter and thicker) and forms chromosomes. So chromatin and chromosomes are the same material in two different states — chromatin is the dispersed form in a non-dividing cell and a chromosome is the condensed form in a dividing cell.

10. Which type of cell is responsible for sending nerve signals?

Answer: Neuron (nerve cell)

Neurons are the specialized cells of the nervous system that transmit messages (nerve impulses) throughout the body. A neuron has a cell body and two types of cytoplasmic extensions: dendrites, which receive impulses and carry them to the cell

body, and an axon, which carries impulses away from the cell body.

11. What do mesophyll cells do in plant leaves?

Mesophyll cells are the green cells present between the upper and lower epidermis of a leaf. They are specialized for photosynthesis: they contain a large number of chloroplasts, which contain the green pigment chlorophyll needed to capture light energy. Their shape and arrangement in the leaf is suitable for maximum absorption of light, and they produce glucose for the plant.

12. How would you define a stem cell?

An unspecialized (undifferentiated) cell that has the ability to make a variety of specialized cell types is called a stem cell. In sexually reproducing organisms all the different types of cells arise from a single cell, the zygote, which is also an unspecialized cell. Stem cells remain in different parts of the body throughout life; they can divide and differentiate into specific cells as the body needs them, and can also regenerate damaged tissue, e.g. stem cells of the skin help in wound healing and those of bone marrow make different types of blood cells.

13. Name the chemical compounds that make up: (a) cell membrane (b) fungal cell wall (c) plant cell wall (d) bacterial cell wall (e) ribosomes (f) chromosomes.

Part	Chemical compounds
(a) Cell membrane	Lipids (phospholipids) and proteins, arranged as a lipid bilayer with embedded proteins; small amount of carbohydrates
(b) Fungal cell wall	Chitin (a polysaccharide)
(c) Plant cell wall	Cellulose and hemicellulose in the primary wall; lignin in the secondary wall
(d) Bacterial cell wall	Peptidoglycan (murein)
(e) Ribosomes	Ribosomal RNA (rRNA) and proteins
(f) Chromosomes	Deoxyribonucleic acid (DNA) and proteins (histones)

14. Label the parts of these cell diagrams.

The diagrams show an animal cell and a plant cell. The parts to be labelled, in each case, are the following:

Animal cell	Plant cell
Cell membrane	Cell wall
Cytoplasm	Cell membrane

Animal cell	Plant cell
Nucleus (with nuclear envelope, nucleolus, chromatin)	Large central vacuole
Mitochondrion	Nucleus (with nuclear envelope, nucleolus, chromatin)
Rough and smooth endoplasmic reticulum	Chloroplast
Golgi apparatus	Mitochondrion
Lysosome	Endoplasmic reticulum
Centriole	Golgi apparatus / cytoplasm

Match these names with the numbers shown on the diagrams of your textbook figure

Remember the difference

• Plant cell has: cell wall, large central vacuole, chloroplast and no centriole (in higher plants).

• Animal cell has: no cell wall, many small vacuoles, no chloroplast, and a pair of centrioles.

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C. Write Answers in Detail

1. Explain the fluid mosaic model of the cell membrane.

Introduction: every cell is covered by a thin, living, partially permeable membrane called the cell membrane. In 1972, Singer and Nicolson gave the fluid mosaic model to explain its structure.

Structure: according to this model the cell membrane is a fluid lipid bilayer with embedded proteins, so it looks like a mosaic of proteins floating in a fluid sea of lipids.

1. Lipid bilayer: the membrane is formed of two layers of phospholipid molecules. Their water-loving (hydrophilic) heads face outward on the two surfaces and their water-hating (hydrophobic) tails face inward, away from water.

2. Proteins: proteins are scattered in the lipid bilayer like tiles in a mosaic. Some proteins are embedded partly in the bilayer (integral proteins) and some lie on the surface (peripheral proteins). Many of them act as channels and carriers for the transport of molecules, and others work as receptors, enzymes or identification marks.

3. Cholesterol: cholesterol molecules lie between the phospholipids and keep the membrane stable and flexible at different temperatures.

4. Carbohydrates: short chains of carbohydrates are attached to the proteins and lipids on the outer surface. They form glycoproteins and glycolipids, which help in cell recognition and communication.

5. Fluidity: the lipid molecules and most proteins can move sideways within the membrane, so the membrane is fluid and not rigid. Because of this fluid nature the cell can change its shape, engulf particles and heal small injuries of its surface.

Conclusion: the fluid mosaic model therefore explains both properties of the membrane: it is fluid (so the cell can move and fuse vesicles) and it is mosaic (because of the proteins and carbohydrates set in the lipid bilayer). It also explains selective permeability — only the substances for which a channel or carrier exists can pass through.

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2. Describe the structure and functions of the cell wall.

Introduction: the cells of bacteria, fungi, plants and some protists (algae) have a rigid, non-living wall outside the cell membrane. It is called the cell wall.

Structure: the plant cell wall is made of three layers:

1. Middle lamella: it is the first layer and is made of calcium pectate; it cements the walls of adjacent cells together.

2. Primary wall: it is present just above the middle lamella and is made of cellulose and hemicellulose; it is thin and elastic so that the young cell can grow.

3. Secondary wall: it is formed on the inner side of the primary wall in mature cells and is made of lignin, which makes it hard and strong.

Chemical nature: the plant cell wall is mainly cellulose. The fungal cell wall is made of chitin and the bacterial cell wall of peptidoglycan. Plasmodesmata — fine

cytoplasmic channels — pass through the plant cell wall and connect the cytoplasm of neighbouring cells.

Functions:

1. It provides shape, strength, protection and support to the inner living matter (protoplasm) of the cell.

2. It protects the cell from mechanical injury and from the entry of harmful organisms.

3. It prevents the cell from bursting when water enters by osmosis, because it can withstand turgor pressure.

4. It gives support to the plant as a whole, so that herbs can stand upright and big trees can carry a heavy trunk and canopy.

5. Its plasmodesmata allow water, minerals and other substances to pass from cell to cell, and it is even used by humans (cotton fibre and wood are plant cell walls).

3. Discuss the components of the nucleus.

Introduction: the nucleus is the largest organelle of a eukaryotic cell. It was observed by Robert Brown in 1831. It is spherical or oval and is surrounded by a double membrane called the nuclear envelope. It controls all the activities of the cell, so it is called the control centre of the cell.

Components of the nucleus:

1. Nuclear envelope: a double membrane that separates the nucleus from the cytoplasm. Its outer membrane continues with the endoplasmic reticulum.

2. Nuclear pores: the envelope has many small pores through which proteins and RNA move between the nucleus and the cytoplasm.

3. Nucleoplasm: the jelly-like fluid present inside the nucleus, in which the chromatin and the nucleolus float.

4. Chromatin: a network of long thread-like fibres made of DNA and proteins. It contains the hereditary information, and it condenses into chromosomes at the time of cell division.

5. Chromosomes: the condensed form of chromatin. Chromosomes are made of deoxyribonucleic acid (DNA) and proteins, and they carry the genes which control the characters of the organism.

6. Nucleolus: a dense, non-membranous body inside the nucleus where ribosomal RNA is made and ribosomes are assembled.

Note: prokaryotic cells do not contain a prominent nucleus; their chromosome is made of DNA only and floats in the cytoplasm. The nucleus is bounded by the nuclear envelope — this is a key difference between eukaryotic and prokaryotic cells.

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4. Describe the structure and function of lysosome and endoplasmic reticulum.

(A) Lysosome — structure: a lysosome is a single membrane-bound sac, formed by the Golgi apparatus, filled with strong digestive (hydrolytic) enzymes. Its inside is acidic. Lysosomes are found in animal cells and in a few plant cells.

Lysosome — functions: (i) they digest waste materials, worn-out organelles and food particles and turn them into simple compounds which the cell can use; (ii) they digest the bacteria and other particles engulfed by the cell, so they protect the cell; (iii) they take part in the digestion of food inside food vacuoles; and (iv) because their enzymes can digest the cell's own parts, they are called the suicidal bags of the cell.

(B) Endoplasmic reticulum — structure: the ER is a network of folded, membrane-bound channels (cisternae) spread throughout the cytoplasm. It is of two types: **rough ER**, which has ribosomes attached to its surface and looks rough, and **smooth ER**, which has no ribosomes and looks smooth. The ER is joined to the outer membrane of the nuclear envelope.

ER — functions: (i) the rough ER synthesizes proteins, which are then sent to the Golgi apparatus; (ii) the smooth ER synthesizes lipids and steroid hormones; (iii) the smooth ER detoxifies harmful substances, specially in liver cells, and stores calcium ions; (iv) the ER provides a large internal surface for chemical reactions and works as the transport system of the cell, carrying materials from one part to another.

5. Describe the formation and function of the Golgi complex.

Discovery: the Golgi apparatus was described by the Italian physician Camillo Golgi in 1898, and it is named after him.

Structure and formation: it is a set of many flattened, membrane-bound sacs (cisternae) stacked over each other, usually present near the nucleus. New cisternae are formed at one side (the forming face) from the membranes of the endoplasmic reticulum, and they mature as they move to the other side (the maturing face), where small vesicles are pinched off. So the Golgi complex is formed continuously from the ER membranes.

Functions:

1. Modification: it modifies the proteins and lipids received from the ER, e.g. by adding carbohydrates (glycosylation).

2. Sorting and packaging: it sorts these materials and packs them into vesicles, each vesicle carrying its material to the right place; in this way it works like the "post office" of the cell.

3. Secretion: the secretory vesicles fuse with the cell membrane and release their contents outside the cell, e.g. digestive enzymes, hormones and mucus.

4. Formation of lysosomes and other vesicles: lysosomes and secretory vesicles are formed from the Golgi membranes.

5. Cell wall material: in plant cells the Golgi complex also makes the materials used in the formation of the cell wall.

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6. Describe the structure and functions of the chloroplast.

Introduction: chloroplasts are the green, double membrane-bound plastids present in the cells of green plants and algae. They are the site of photosynthesis and they contain the green pigment chlorophyll.

Structure:

1. Chloroplast envelope: it is bounded by two membranes — the outer and the inner membrane.

2. Stroma: the jelly-like fluid present inside the chloroplast. It contains enzymes for photosynthesis, ribosomes, DNA and starch granules.

3. Thylakoids: the inner membrane forms a system of flattened sacs called thylakoids. The pigment chlorophyll is present on the membranes of the thylakoids.

4. Grana: the thylakoids are stacked over each other in groups, like piles of coins; each stack is called a granum. The grana are joined to one another by lamellae.

Functions: (i) chloroplasts trap light energy and prepare glucose from carbon dioxide and water by photosynthesis; (ii) they release oxygen into the air; (iii) they store the food as starch and also contain their own DNA, so they can make some of their own proteins and can increase in number. Other plastids are chromoplasts (coloured, give colour to flowers and fruits) and leucoplasts (colourless, store starch, oil or proteins).

7. How does turgor pressure develop in a plant cell?

Definition: the outward pressure of the central vacuole on the cytoplasm and cell wall of a plant cell is called turgor pressure, and the condition is called turgor.

How it develops: a mature plant cell has a single large central vacuole filled with cell sap — a watery solution of salts and sugars. When water enters the cell by osmosis (because the sap is more concentrated than the surrounding water), it collects in this vacuole. The vacuole grows and pushes outward against the cytoplasm, and the cytoplasm in turn pushes against the strong cell wall.

Result: this outward pushing force is the turgor pressure. The cell wall can resist it, so the cell does not burst; instead it becomes firm (turgid). The cell wall pushes back with an equal and opposite pressure called wall pressure. Turgor pressure:

1. keeps the leaf and stem cells firm, which is why soft plant parts stand upright (this is the cause of wilting when water is short);

2. maintains the shape of the cell and of the plant;

3. helps in the opening and closing of stomata (guard cells become turgid and the stoma opens); and

4. drives the growth of cells, because the pressure stretches the elastic primary wall.

8. Write any four differences between a plant cell and an animal cell.

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Plant cell	Animal cell
It has a rigid cell wall made of cellulose outside the cell membrane.	It has no cell wall; only a flexible cell membrane is present.
It has a single large central vacuole with cell sap, which provides turgor pressure.	It has many small, temporary vacuoles, or none.
It contains chloroplasts, so it can prepare its own food by photosynthesis.	It has no chloroplast, so it cannot photosynthesize.
Higher plants have no centrioles; the shape is fixed by the cell wall.	It has a pair of centrioles that form spindle fibres during cell division.
Nucleus is pushed to one side by the large vacuole; the cell is usually regular in shape.	Nucleus lies in the centre; the cell usually has an irregular shape.

9. Describe the concept of division of labour and how it applies in multicellular organisms. Give at least three examples.

Concept: division of labour means the specialization of different parts of a system to perform specific tasks more efficiently. It is a fundamental principle of all biological systems, both within and across cells: when every part does one job well, the whole system works better than if every part tried to do every job.

Within a cell: the organelles each carry out distinct functions necessary for the cell's survival. Mitochondria generate energy, endoplasmic reticulum synthesizes proteins and lipids, and lysosomes break down waste materials. In this way the function of each organelle contributes to the cell's overall survival, growth and functioning. The division of labour explains why a eukaryotic cell has many separate membrane-bound organelles instead of one big compartment.

Across cells: in multicellular organisms each type of cell performs a specific role and contributes to the functions of the whole organism. This intercellular specialization allows complex organisms to perform a wide range of functions.

Examples:

1. Muscle cells are specialized for contraction and movement; they contain actin and other contractile proteins, so the whole body can move and the heart can pump blood.

2. Nerve cells (neurons) are specialized for transmitting messages; their long axon and branched dendrites carry impulses quickly between the brain and other organs.

3. Red blood cells are specialized for carrying oxygen; they are filled with haemoglobin and have no nucleus, so they can carry the maximum oxygen from the lungs to the tissues.

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4. Mesophyll cells of the leaf are specialized for photosynthesis and epidermal cells for protection; root hair cells absorb water, and guard cells control the stomata.
Conclusion: division of labour increases efficiency and functionality; it is the reason why a single zygote can build up tissues, organs and organ systems of a complex organism.

10. Write a note on cell specialization.

Definition: in multicellular organisms there are different types of cells. Each type has a special structure and performs a special function. When cells are formed by cell division they are all similar; after their formation they undergo the process of specialization or differentiation. During this process they get special sizes, shapes, structures and metabolic features, and as a result they become specialized.
Specialized cells of plants:
1. Mesophyll cells: green cells of the leaf with a large number of chloroplasts; specialized for photosynthesis because their shape and arrangement is suitable for maximum absorption of light.
2. Epidermal cells: flat, tightly packed cells that make the outer layer (epidermis) of plant organs and protect the internal tissues. Modified epidermal cells do other jobs: root hair cells increase the surface area for absorption of water and minerals, and guard cells regulate the opening and closing of stomata.
Specialized cells of animals:
3. Muscle cells: elongated cells filled with actin and other contractile proteins; they contract to produce movement. Skeletal muscle cells are long, striated and voluntary; cardiac muscle cells are branched, striated and involuntary; smooth muscle cells are spindle-shaped, non-striated and involuntary.
4. Neurons: cells of the nervous system with a cell body, dendrites and axon; they transmit nerve impulses throughout the body.
5. Red blood cells: biconcave disc-shaped cells filled with haemoglobin; they carry oxygen from the lungs to the tissues. In mammals the mature red blood cell has no nucleus, mitochondria or ER, so it can hold more haemoglobin.
6. Liver cells (hepatocytes): specialized for storage of glycogen, iron and some vitamins, detoxification of toxic substances, production of blood-clotting proteins and recycling of old red blood cells. They have prominent nuclei, many mitochondria, an expansive SER and many peroxisomes for these jobs.
7. Other examples: white blood cells (defence), platelets (clotting), sperm and egg cells (reproduction), and nerve and fat cells — each with a structure that fits its function.
Importance: cell specialization is the base of division of labour in the body. Because each cell type is perfect for one job, a multicellular organism can perform many different functions at the same time and can grow much larger and more complex than a single cell.

D. Inquisitive Questions

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1. What impact might mitochondrial dysfunction or absence have on other organelles' ability to operate in a cell?

1. No energy supply: mitochondria produce ATP by cellular respiration. Almost every other organelle needs ATP, so without mitochondria the ribosomes could not join amino acids, the ER and Golgi could not make and transport proteins, and the pumps of the cell membrane would stop.

2. Ribosomes and protein synthesis would stop: protein synthesis is an energy-consuming process; without ATP the cell cannot make enzymes, and without enzymes all metabolism stops.

3. Active transport would fail: the cell membrane could no longer move ions and nutrients against their concentration gradient, so the internal environment of the cell (pH, ion balance) would be lost.

4. Lysosomes and digestion would be affected: the digestive enzymes and the pumping of hydrogen ions into lysosomes need energy; waste would collect in the cell.

5. Transport inside the cell would stop: movement of vesicles along microtubules (from ER to Golgi to membrane) needs ATP, so secretion would stop and materials would pile up.

6. Cell division would stop: the spindle fibres need energy, so the cell could not divide or repair itself; ultimately the cell would die and, in a multicellular organism, the tissue would fail.

Conclusion: mitochondria are not just one organelle among many; they are the energy centre on which the working of all the other organelles depends.

2. What may happen if the coordination between the ribosomes and the nucleus were to fail, and why is it so important?

What could happen: the nucleus (through its DNA) carries the instructions for making proteins, and the ribosomes in the cytoplasm carry out those instructions. If this coordination fails:

1. Wrong proteins would be made: the ribosome would translate a wrong or damaged message, so useless or harmful proteins would be produced and the cell's enzymes would not work.

2. Protein synthesis would stop: if mRNA cannot leave the nucleus through the nuclear pores, or if the ribosomes do not receive the message, no protein is made and the cell cannot grow, repair or divide.

3. Ribosomes would not be formed: the ribosomal RNA and the ribosomal subunits are made in the nucleolus; without the nucleus no new ribosomes can be manufactured.

4. The cell would lose control: the proteins include the enzymes and the regulatory proteins that control every activity; without them the cell's metabolism would collapse and the cell would die. In a body, such failure appears as serious diseases (for example uncontrolled cell division leading to cancer).

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Why it is important: this coordination is the central link of gene expression — DNA — RNA — protein. The nucleus stores and protects the hereditary information, and the ribosome is the only place where that information is turned into working proteins. Keeping them coordinated also protects the DNA (which stays safe inside the nucleus) while proteins are made in the cytoplasm. That is why this communication through the nuclear pores is one of the most important controls of the cell.

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Points to Remember

The cell is the fundamental building block of life. Robert Hooke discovered it in 1665; Robert Brown observed the nucleus in 1831. There are two basic types of cells: prokaryotic (simple, no membrane-bound organelles) and eukaryotic (complex, with membrane-bound organelles).

Cell wall: rigid, non-living layer outside the cell membrane of bacteria, fungi, plants and some protists. Primary wall = cellulose and hemicellulose; secondary wall = lignin; fungal wall = chitin; bacterial wall = peptidoglycan.

Cell membrane: made of a lipid bilayer with embedded proteins (fluid mosaic model); it is partially permeable and controls the movement of substances.

Cytoplasm is the jelly-like substance between the nucleus and the cell membrane; it holds the organelles and is the site of many chemical reactions.

Nucleus: spherical or oval, surrounded by a double nuclear envelope with pores; contains nucleoplasm, chromatin, chromosomes and nucleolus. Chromosomes are made of DNA and proteins and carry the hereditary information.

Cytoskeleton: a network of microtubules, microfilaments and intermediate filaments; it maintains cell shape, holds organelles in place and makes spindle fibres, cilia and flagella.

Ribosomes: tiny granular structures made of rRNA and proteins, without a membrane; they are the site of protein synthesis and are either free in the cytoplasm or attached to the rough ER.

Endoplasmic reticulum: network of membrane-bound channels. Rough ER (with ribosomes) makes proteins; smooth ER makes lipids, detoxifies harmful substances and stores calcium.

Golgi apparatus: a set of many flattened sacs (cisternae) stacked over each other; it modifies, sorts, packages and secretes proteins, and forms lysosomes.

Lysosomes: single membrane-bound sacs with strong digestive enzymes that break down biomolecules, waste and worn-out organelles (suicidal bags of the cell).

Vacuoles: membrane-bound fluid-filled sacs. A mature plant cell has one large central vacuole whose outward pressure (turgor pressure) keeps the cell firm.

Mitochondria: the "powerhouses" of the cell; double membrane with cristae; they produce energy (ATP) by cellular respiration and can double their number by themselves because they have their own DNA.

Chloroplasts: responsible for photosynthesis; they have a double membrane, stroma, thylakoids and grana containing chlorophyll.

Centriole: barrel-shaped, formed of 9 groups of microtubule triplets; found in animal cells and most protists, it forms spindle fibres during cell division.

Specialized cells: mesophyll (photosynthesis) and epidermal (protection) cells in plants; muscle (movement), neuron (nerve impulses), red blood (oxygen transport) and liver (storage and detoxification) cells in animals.

Division of labour: different parts of a system are specialized to perform specific tasks more efficiently — it works within a cell (organelles) and across cells (tissues and organs).

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Stem cells: undifferentiated or unspecialized cells that can differentiate into specific cells; they repair and regenerate tissues (skin, liver, bone marrow).

Part	Main function
Cell membrane	Boundary; controls the movement of substances in and out of the cell
Cell wall	Shape, strength, protection and support
Nucleus	Control centre; stores hereditary information
Ribosome	Protein synthesis
Rough ER	Synthesis and transport of proteins
Smooth ER	Synthesis of lipids; detoxification; calcium storage
Golgi apparatus	Modification, packaging and secretion of proteins
Lysosome	Digestion of waste and worn-out organelles
Mitochondrion	Energy production by cellular respiration
Chloroplast	Photosynthesis
Vacuole	Storage of water and food; turgor pressure
Centriole	Spindle formation in cell division

Table: Organelles and their main functions — quick revision

How to attempt these questions in the paper

• MCQ: write only the correct option (a, b, c or d).

• Short question: 2 to 4 sentences; if the question asks for a function, start with the name of the organelle.

• Difference question: always draw a two-column table with headings.

• Detailed answer: give the structure first and then the functions as numbered points. Diagrams of the nucleus, mitochondrion and chloroplast carry extra marks.